
A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived
analysis

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Abstract

1 Tensor decomposition factors a large table of interactions into a small set of low-
2 rank components that can be ranked and inspected. Applying it directly to a
3 conventional robot policy is difficult because softmax, ordinary activations, and
4 normalization interrupt the required algebraic form. We introduce χ -VLA, a com-
5 pact vision-language-action policy whose learned map uses only bilinear, linear,
6 and rational operations. Its checkpoint therefore defines an explicit rational tensor
7 program, making interactions among image, instruction, state, and action coordi-
8 nates accessible from the weights. A mechanical audit finds no hidden incompatible
9 operation, and local reconstructions agree near the float32 precision floor. On all
10 ten LIBERO-Object tasks, the convolutional 20.1M-parameter instantiation reaches
11 $93.7\% \pm 0.8\%$ under the published three-seed protocol. Averaging its three fixed
12 checkpoints reaches 96.2% at three times the inference cost. In a separate three-
13 checkpoint ViT instantiation, we derive an exact weight-based decomposition of
14 softmax-free attention and reconstruct every attention module with worst relative
15 error 2.56×10^{-7} . All 36 primary block-head tests favor its rank-128 subspace
16 over equal-rank random controls, with median fidelity ratio 2.44. The evidence es-
17 tablishes strong specialist capability and exact layerwise access to trained attention
18 coefficients. It does not establish broad robotic generalization, robust instruction
19 grounding, or an exact end-to-end decomposition of the full policy.

20 1 Introduction

21 Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains
22 millions of learned numbers, but it does not directly reveal which combinations of image regions,
23 instruction tokens, and robot state produce an action. Analysts therefore usually collect activations
24 on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the
25 same structure is present in the weights and whether it causes behavior.

26 Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a
27 decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization
28 finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios
29 of polynomials, its weights define an explicit table of input interactions. Decomposing that table can
30 rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that
31 every learned block preserves this algebraic form. The aim is not merely compression. It is a direct
32 path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

75 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 76 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

77 Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a
 78 numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization
 79 operations update this pair. The model therefore keeps input-specific rescaling while remaining
 80 exactly representable as a rational tensor network.

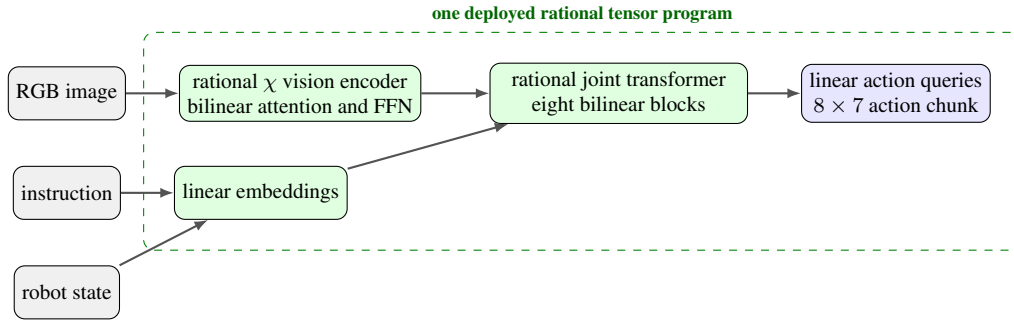


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

81 2.2 Architecture and training

82 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 83 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 84 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 85 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 86 continuous action dimensions. The complete model has 20,137,352 learned parameters.

87 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 88 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 89 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 90 that verify every requested state was loaded.

91 2.3 Mechanical certificate on the trained checkpoint

92 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 93 point at runtime and reconstruct representative operations from saved weights.

Mechanical decomposability audit of the trained 189/200 checkpoint

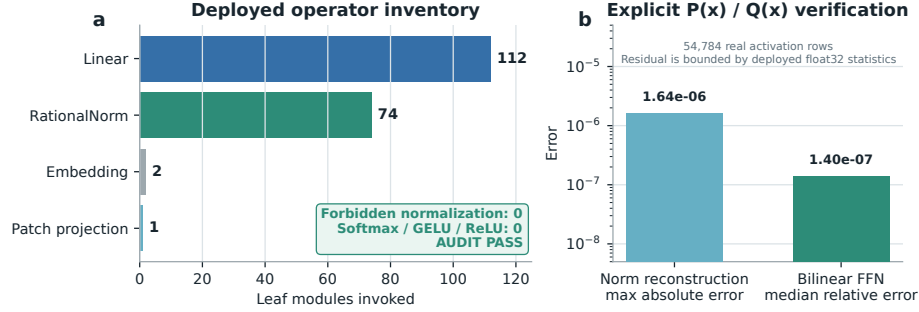


Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

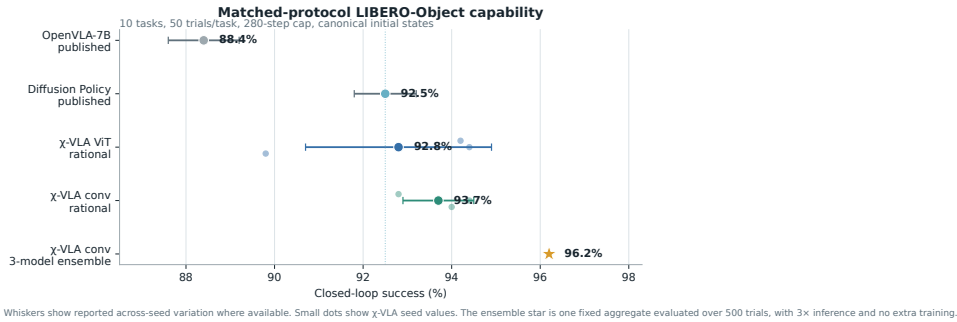


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

107 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

108 The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high
 109 specialist capability is reproduced across three independently trained checkpoints. These are historical
 110 Modal aggregate records. A locked replay is producing immutable episode-level outcomes.

111 Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and
 112 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points
 113 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato
 114 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one
 115 20.1M-parameter policy.

116 3.3 What this result does and does not establish

117 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
 118 existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad
 119 robotic generalization or state of the art.

120 4 Exact weight-derived structure through attention

121 4.1 Exact layerwise construction

122 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
 123 a high-order coefficient table. Each entry says how strongly one combination of query, key, and
 124 value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe
 125 are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and
 126 an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to
 127 1.24×10^{-14} .

128 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 129 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 130 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 131 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 132 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
 133 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
 134 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
 135 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

136 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 137 values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays
 138 the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most
 139 coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force
 140 materialization to 2.13×10^{-14} .

141 4.2 Trained policy result

142 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
 143 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
 144 sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace.

145 Real cached activations are used only afterward to measure how faithfully that subspace preserves
 146 the layer’s computation.

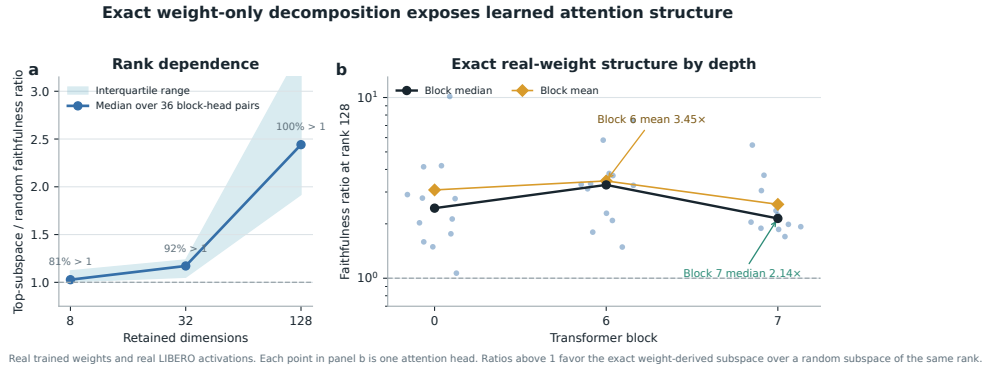


Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

147 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
 148 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
 149 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
 150 0, 6, and 7.

151 This extends exact weight-only construction through individual attention layers. It is not an exact
 152 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
 153 rapid degree and representation growth.

154 5 Related work

155 **VLA capability.** OpenVLA established an open pretrained VLA and standardized LIBERO com-
 156 parisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can
 157 substantially improve adaptation and control efficiency [2]. These systems use large pretrained
 158 backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

159 **Tensor and polynomial networks.** Polynomial networks and tensor networks use explicit mul-
 160 tilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight
 161 eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for com-
 162 positional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an
 163 exact reduced construction for individual softmax-free attention layers.

164 **Causal mechanisms and policy analysis.** Bilinear IMPALA policies have combined competitive
 165 ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5].
 166 This is the closest architectural precedent for weight-based policy analysis. Recent work steers
 167 pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify
 168 and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions
 169 recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These
 170 results make a universal separation from conventional policies untenable. Our intended distinction is
 171 empirical: exact weight auditing in a strong continuous vision-language policy.

172 **VLA robustness.** Recent evaluations show that standard LIBERO success can coexist with weak
 173 robustness and language use [9–11]. We therefore treat LIBERO-Object as a bounded specialist
 174 capability setting rather than evidence of broad generalization.

6 Limitations and conclusion

χ -VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis now crosses individual attention layers.

The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions, so these results do not establish broad robotic generalization or robust instruction grounding. Exact reconstruction is certified layer by layer and does not yet yield one compact symbolic contraction of the full policy. The three-checkpoint ensemble trades three forward passes for its strongest capability result.

Within that scope, polynomial and rational restrictions are compatible with strong closed-loop control and provide a precise weight-derived analysis surface. Convolutional checkpoints support the capability result, while separate ViT checkpoints support the exact reconstruction. We do not claim a direct coefficient-edit interface.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

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219 A Supporting results and decisions

220 A.1 Per-task ensemble behavior

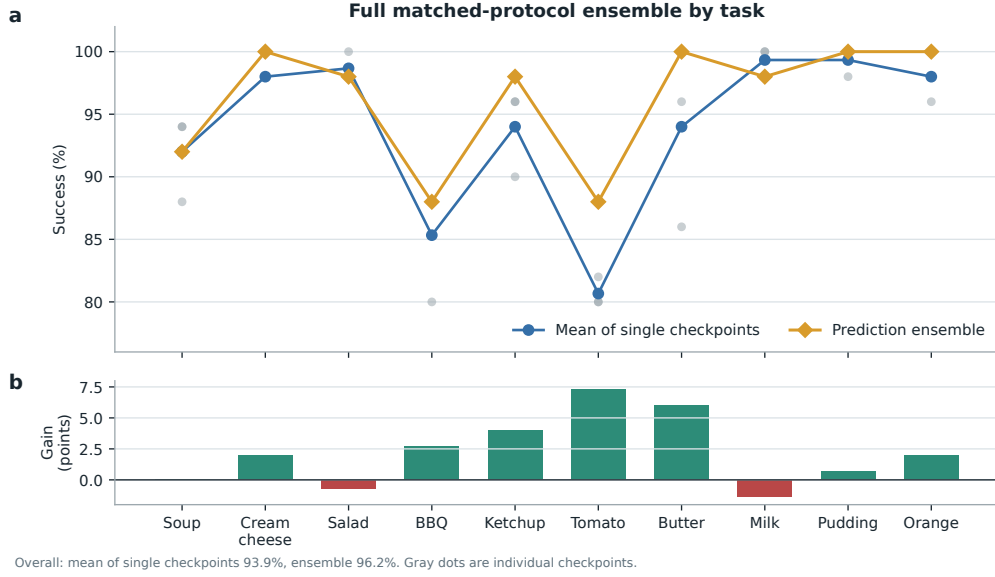


Figure 5: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

221 A.2 Expanded attention screen

222 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 223 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 224 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
 225 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
 226 tested rank.

227 A.3 Numerical scope of rational conversion

228 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 229 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 230 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 231 establish practical numerical safety under distribution shift.